

Notes: Darmouni & Siani (JME, 2025)

Bond market stimulus: Firm-level evidence

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1 Big picture

- **Question.** When the Federal Reserve supported the U.S. corporate bond market in spring 2020, what did issuing firms actually do with the funds? Did cheaper bond financing translate into real investment, or mainly into balance-sheet reshuffling?
- **Setting.** The paper studies the March 23 and April 9, 2020 Federal Reserve announcements creating corporate bond purchase facilities during the COVID market panic. It tracks corporate bond issuers using micro-data on bond issuance, balance sheets, bank debt, credit lines, and equity payouts.
- **Core empirical challenge.** Bond issuance surged after the policy announcements, but the pandemic itself was simultaneously a huge shock to cash flows, uncertainty, and investment opportunities. A simple before-versus-after comparison cannot isolate the effect of the policy from the effect of COVID.
- **Main conceptual contribution.** The paper shows that unconventional monetary policy aimed at bond markets can transmit mainly through *corporate liquidity management* rather than through immediate real investment. Firms borrowed heavily, but much of the borrowing financed cash accumulation, debt substitution, and rollover rather than new capital spending.
- **Main empirical design.** The paper combines four pieces of evidence:
 - within-firm event studies around bond issuance, comparing spring 2020 with normal times;
 - an instrumental-variables design using each investment-grade issuer’s ex ante weight in the Fed’s market-portfolio rule;
 - a high-frequency design using announcement-driven credit-spread changes tied to bond-purchase eligibility;
 - direct evidence on bank credit lines, loan repayment, and share repurchases.
- **Main baseline result.** After the intervention, issuing firms accumulated large and persistent cash balances. Unlike in normal times, that cash stayed high for many quarters, even up to two years after issuance.
- **Main real-side result.** Real investment was weak. Property, plant, and equipment did not rise meaningfully after issuance, and the causal designs show little evidence that more policy-exposed firms invested more.
- **Main causal result.** Firms that were more exposed to the Fed’s intervention because of index weight or announcement-induced spread compression became more likely to issue and to hold more cash, but not more likely to raise investment. The paper also reports only suggestive and limited employment effects.
- **Key financial-heterogeneity result.** Many issuers already had substantial unused bank liquidity. Among firms issuing bonds after March 23, 2020, large fractions of both high-yield and investment-grade firms left credit lines undrawn, while others drew first and then quickly repaid banks after issuing bonds.

- **Key payout result.** Equity payouts fell somewhat after issuance, but they did not disappear. Nearly half of issuers still repurchased shares in spring 2020, which suggests that many issuers did not place an extremely high marginal value on inside liquidity.
- **Use-of-funds decomposition.** Out of each dollar of post-intervention bond issuance, roughly 45 cents went to cash accumulation, 15 cents to paying down bank debt, and 30 cents to rolling over existing bonds, with essentially zero going to real investment on average.
- **Economic takeaway.** Bond-market support revived issuance, but its firm-level transmission was mainly a *liquidity and capital-structure* channel, not a strong investment channel. The paper’s central message is that the macro effects of unconventional policy depend heavily on the marginal value of liquidity for the firms being targeted.

2 Data and measurement (institutional context and key objects)

2.1 Why the 2020 intervention is the right laboratory

- Early 2020 combined a large real shock with severe corporate-bond-market dislocation.
- Bond spreads spiked, secondary-market functioning deteriorated, and issuance briefly collapsed, especially for riskier firms.
- On March 23 and April 9, 2020, the Federal Reserve announced unprecedented support for corporate bond markets.
- The market response was immediate: issuance rebounded sharply, even though actual Fed purchases later turned out to be small relative to the overall market.

Interpretation. This episode is useful because the policy shock was both large and highly visible. The announcements clearly changed financing conditions in corporate bond markets, which lets the paper ask a sharp question: once the market reopened, what did firms do with the money?

2.2 Policy timeline and eligibility

- The first key announcement is **March 23, 2020**, when the Fed announced support for investment-grade corporate bonds and ETFs.
- The second is **April 9, 2020**, when the program size expanded and eligibility broadened, including some high-yield exposure through ETFs and recently downgraded “fallen angel” firms.
- For the high-frequency identification, the important eligibility margins are whether a firm’s bonds were investment-grade, had remaining maturity below five years, and were in ETF-eligible segments.

Interpretation. The paper exploits both the broad market rebound after the announcements and

the cross-sectional fact that some issuers were more directly favored by the program’s purchase rules.

2.3 Core data sources

- **Mergent FISD**: security-level corporate bond issuance data.
- **Compustat**: quarterly firm balance-sheet and operating variables.
- **Capital IQ**: quarterly debt-composition data, including revolving credit, total bank debt, and undrawn revolving credit.
- **Enhanced TRACE**: daily corporate bond transaction data used to construct firm-level spread changes for the high-frequency design.

Interpretation. The key strength is not just seeing issuance. The paper can connect issuance to subsequent cash, investment, debt composition, and payout behavior.

2.4 Sample construction

- The sample includes U.S. dollar corporate bonds with at least **\$100 million** face value issued by firms reporting in U.S. dollars.
- The paper excludes financials, sovereigns, utilities, convertibles, capital impact bonds, community bonds, PIK securities, exchanged bonds, and some transaction-specific bonds tied to the T-Mobile/Sprint acquisition.
- The main spring 2020 “post-Fed” window is **March 23 to June 30, 2020**.
- The “normal” comparison period is **2010–2019**.
- The panel follows 2020 issuers’ balance sheets through **mid-2022**, up to two years after first issuance.

Interpretation. The design deliberately compares the unusual post-announcement issuance wave to a long pre-COVID period in which firms also issued bonds, but without the same policy environment.

2.5 Sample size and descriptive scale

- In the post-Fed window, the matched sample contains **317 firms** issuing **598 bonds**.
- In the normal period, it contains **1297 firms** issuing **6645 bonds**.
- The paper matches roughly **84%** of bonds to Compustat firms and keeps the subset with compatible firm-level financial reporting for balance-sheet analyses.
- During March–May 2020, the median bond in the sample is about **\$650 million**, with median maturity around **8 years** and yield about **4.03%**.

- For March–May 2017–2019, the comparable median bond is about **\$500 million**, also with roughly **8 years** maturity and yield about **4.88%**.

Interpretation. Issuance in 2020 was not only more intense; it also took place at materially lower yields after the policy intervention.

2.6 Main balance-sheet and financing variables

- **Cash** is cash plus short-term investments, scaled by either 2019 year-end assets or lagged assets depending on the figure.
- **Real investment** is primarily proxied by property, plant, and equipment (PPE); the paper also considers non-cash assets in robustness checks.
- **Bank liquidity** comes from revolving credit, total bank debt, and undrawn revolving credit from Capital IQ.
- **Equity payouts** are measured with a share-repurchase dummy rather than ordinary dividends, because repurchases are more discretionary.
- **Abnormal issuance** equals issuance in March 23–June 30, 2020 minus the firm’s average quarterly issuance over 2000–2019.
- **Exposure to the intervention** is measured either by the firm’s Fed market-portfolio index weight or by predicted announcement-induced spread compression.

Interpretation. The paper’s central outcomes are chosen to distinguish *real use of funds* from *liquidity management and refinancing*.

2.7 Rating groups and the paper’s main heterogeneity split

- **Investment grade (IG)** firms are rated BBB- or above.
- The paper often separates BBB-rated issuers from safer **A or above** issuers.
- **High yield (HY)** firms are rated below BBB-.

Interpretation. This rating split matters because the Fed targeted investment-grade names more directly, but the paper also wants to know whether the intervention’s market rebound indirectly relaxed financing for riskier firms.

2.8 Descriptive facts that motivate the paper

- High-yield firms issued over **\$120 billion** and investment-grade firms over **\$500 billion** in January–May 2020, despite the brief issuance freeze in March.
- The introduction emphasizes that, unlike normal times, as much as **90 cents of each dollar issued** went on average to increasing cash or repaying existing debt, with essentially **zero increase in real investment**.

- The bank-credit data show that a surprisingly large set of issuers had unused committed liquidity from banks before or during the post-announcement issuance wave.

Interpretation. The question is not whether the intervention revived markets; it clearly did. The real question is how revived bond markets translated into firm behavior.

3 Models and empirical specifications (all equations + interpretation)

3.1 Dynamic event study around bond issuance

The paper’s baseline balance-sheet event study is

$$Y_{fq} = \sum_{m=-4}^8 \beta_m Issue_{f,q-m} + \alpha_f + \alpha_{ind \times year} + \epsilon_{fq}. \quad (1)$$

Here:

- Y_{fq} is a firm-quarter outcome such as cash/assets, PPE/assets, or a repurchase dummy,
- $Issue_{f,q-m}$ indicates the quarter relative to the firm’s issuance event,
- α_f are firm fixed effects,
- $\alpha_{ind \times year}$ are industry-year fixed effects.

The regression is run separately for normal times and for bonds issued between March 23 and June 30, 2020.

Interpretation. This specification asks: how do a firm’s balance-sheet outcomes move in the quarters before and after issuance? The fixed effects mean the paper is comparing each firm mostly to itself, while also absorbing broad industry shocks in a given year. The point is not just whether issuers hold more cash in 2020 than in 2019, but whether the *dynamic response to issuance* looks different in spring 2020 from normal issuance episodes.

3.2 Index-weight IV: first stage

The first causal design uses the firm’s Fed market-portfolio weight as an instrument for abnormal issuance:

$$AbnormalAmtIssued_{f,2020H1} = \gamma WeightIndex_f + \beta X_f + \alpha_{ind} + \epsilon_f. \quad (2)$$

The dependent variable is the amount issued in March 23–June 30, 2020 minus the firm’s average quarterly issuance from 2000–2019. The control vector X_f includes rating, 2019 firm size, cash, leverage, sector, and Tobin’s Q .

Interpretation. The idea is that the Fed said it would buy a market portfolio of eligible bonds. Firms with larger index weights were therefore more exposed to the policy support, even after

controlling for observable firm characteristics. A strong first stage means those more exposed firms indeed issued more than usual.

3.3 Index-weight IV: second stage

The second stage interacts the predicted issuance with quarter dummies:

$$\ln(Y)_{fq} = \sum_{m=-4}^8 \beta_m \mathbf{1}\{q = 2020Q1 + m\} \times \widehat{AbnormalAmtIssued}_f + \alpha_f + \epsilon_{fq}. \quad (3)$$

The main outcomes are log cash and log real investment; the paper also adapts the design to employment.

Interpretation. This is a dynamic causal event study. Instead of asking what happened to firms that issued, it asks what happened to firms that were *induced by policy exposure* to issue more. That helps separate the effect of the intervention from the broader macro shock.

3.4 High-frequency announcement design

The second causal design starts from credit-spread movements around the March 23 and April 9 announcements. In words, the paper regresses firm-level cumulative spread changes on announcement-window indicators interacted with bond-purchase eligibility: whether the firm is investment-grade, how much of its debt is short maturity, and how much sits in ETF-eligible segments. Using those coefficients, it constructs a *predicted policy-driven spread decline* for each firm.

The balance-sheet regression is then

$$\ln(Y)_{fq} = \sum_{m=2019Q2}^{2022Q2} \beta_m \mathbf{1}\{q = m\} \times \mathbf{1}\{HighExposure_f\} + \alpha_f (+\alpha_q) + \epsilon_{fq}, \quad (4)$$

where $HighExposure_f = 1$ for firms in the top quartile of predicted announcement-induced spread declines.

Interpretation. This design uses the fact that the announcements helped some firms' bonds more than others for policy-specific reasons. If those firms later issued more and held more cash, the case that the intervention itself mattered becomes stronger.

3.5 Bank-credit and payout analyses

For bank credit, the paper mainly uses quarter-to-quarter changes in Capital IQ debt stocks to approximate flows in revolving credit and total bank debt. It classifies firms by how much of their available end-2019 credit line they drew in 2020Q1 and then studies how much of that borrowing they repaid after issuing bonds in 2020Q2.

For equity payouts, the paper returns to the dynamic event-study specification above but changes the dependent variable to an indicator for positive share repurchases in the quarter.

Interpretation. These analyses are crucial because they tell us whether bond issuance relaxed a binding liquidity constraint or whether it mostly replaced other financing sources and coexisted with payouts to shareholders.

3.6 Use-of-funds decomposition

To map each dollar of bond proceeds into uses, the paper estimates

$$\frac{Y_{fq}}{Assets_{2019_f}} = \sum_{m=-4}^4 \beta_m \frac{AmtIssued_{f,q-m}}{Assets_{2019_f}} + \alpha_f + \alpha_{ind,y} + \frac{CFO_{fq}}{Assets_{2019_f}} + \epsilon_{fq}. \quad (5)$$

The dependent variable Y_{fq} is alternately cash, real investment, bank debt, rollover bonds, or payouts to equity. The paper then focuses on $\beta_{t+2} - \beta_{t-2}$ to summarize how the balance sheet shifts around issuance.

Interpretation. Because both issuance and outcomes are normalized by 2019 assets, the coefficients can be read as a *dollar decomposition*. This is what lets the paper say how much of each dollar of issuance went to cash, bank-debt repayment, bond rollover, and real investment.

3.7 Estimation details that matter

- Standard errors are generally clustered at the **2-digit industry** level.
- Ratios are typically winsorized at the **1%** level.
- The causal IV design is limited to **investment-grade** issuers because they are the firms directly targeted by the Fed purchase rule.
- The high-frequency spread design uses data from roughly **March 20 to April 13, 2020** to isolate the announcement effect.

Interpretation. These choices are important because the paper is working with a single extraordinary episode, so sensible clustering, trimming, and tight event windows matter a lot for credibility.

4 Identification and empirical design (what makes the estimates credible)

4.1 What is descriptive and what is causal

- The baseline event studies are highly informative about how issuers behaved, but they are not by themselves causal because the pandemic and the intervention happen at the same time.
- The paper is explicit about this limitation and then adds two separate causal designs.

Interpretation. The paper does not oversell the descriptive event studies. Instead, it uses them to establish the main facts and then asks whether more policy-exposed firms behaved differently.

4.2 Why the index-weight IV is plausible

- The instrument comes from a pre-announced Fed rule: purchases were designed to mirror the market portfolio of eligible corporate bonds.
- Conditional on controls, a firm's market-portfolio weight should not directly measure how hard the pandemic hit that firm.
- The paper controls for rating, size, leverage, cash, sector, and Tobin's Q , and includes industry fixed effects in the first stage.
- The paper also notes that index weight is uncorrelated with an external measure of COVID exposure based on industry abnormal employment decline in 2020Q1.

Interpretation. The instrument is attractive because it is rooted in the policy rule itself rather than in firms' own financing choices. The key identifying assumption is that, after conditioning on observables, market weight affects later cash and investment mainly by changing policy-induced issuance.

4.3 Why the high-frequency spread design is useful

- Bond-purchase eligibility created sharp cross-sectional differences in announcement-day spread movements.
- The spread response is measured in a narrow window centered on the actual policy announcements.
- The paper first shows that larger predicted spread declines made firms more likely to issue; it then studies balance-sheet responses for the firms with the largest predicted treatment.

Interpretation. This design is helpful because it leans on very short-run market reactions that are hard to explain with slow-moving firm fundamentals. It is a different source of variation from the index-weight IV, so similar results across the two designs matter a lot.

4.4 Why the bank-credit evidence is central to interpretation

- If firms that issued bonds also had large unused credit lines, then weak investment cannot simply be read as evidence that financing conditions still stayed too tight.
- If firms issued bonds and then repaid bank loans, a large part of the effect is debt substitution rather than new spending.
- If firms kept repurchasing shares, then some issuers evidently did not value each marginal dollar of liquidity very highly.

Interpretation. These facts do not provide a separate instrument, but they are powerful for rejecting overly simple stories in which spring 2020 bond issuance mechanically translated into higher investment because firms were starving for funds.

4.5 Alternative explanations the paper considers

- **The pandemic reduced investment demand anyway.** True, and this is exactly why the paper adds causal designs comparing more versus less policy-exposed firms.
- **Bonds just became much cheaper than loans.** The paper argues this is incomplete: loan rates also fell, and the bond-loan spread did not clearly compress enough to explain everything.
- **Banks may have restricted undrawn lines.** The paper argues against this using broad evidence of healthy bank balance sheets, strong line honoring, limited covenant enforcement, and the fact that riskier issuers actually drew more.

Interpretation. The paper’s preferred story is not simply “bonds were cheap.” It is that the intervention made long-term market financing available, and firms used that access to build liquidity buffers and reshape liabilities.

4.6 What the paper can and cannot say

- The paper can say that average firm-level transmission to cash was strong and transmission to real investment was weak.
- The paper can say that many issuers had alternative liquidity and that a significant fraction of bond proceeds went to debt substitution and rollover.
- The paper cannot fully estimate the aggregate counterfactual without the intervention.
- The paper cannot rule out that the intervention prevented a much worse financial meltdown or indirectly helped other firms through the banking system.

Interpretation. This is a firm-level transmission paper, not a full welfare or general-equilibrium policy evaluation.

4.7 Relation to the literature and why the result is interesting

- The findings challenge models in which firms mainly borrow to invest and support models in which firms borrow to hold liquidity when funding conditions temporarily improve.
- Relative to evidence on the ECB’s Corporate Sector Purchase Programme, the U.S. 2020 intervention appears to have worked through a different balance-sheet channel.
- The paper argues that macro models of monetary transmission should explicitly include cash, long-term debt, and multiple financing sources.

Interpretation. The big contribution is not just another reduced-form estimate. It is a micro-foundational message about what kinds of firm-financing margins monetary models need to include.

5 Tables and figures

5.1 Figure 1: cash and investment around issuance

Read.

- For investment-grade issuers in normal times, cash rises modestly around issuance and then mean-reverts within a couple of quarters.
- For investment-grade issuers after March 23, 2020, cash rises sharply and remains elevated for many quarters, even up to two years.
- PPE does not show a comparable increase after post-Fed issuance.
- The paper reports similar qualitative patterns for high-yield issuers in the appendix.

Economic message. Spring 2020 bond issuance looked very different from ordinary issuance episodes. Firms were not mainly borrowing to expand their capital stock; they were borrowing to sit on liquidity.

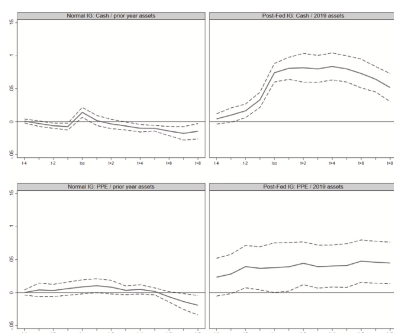


Fig. 1. Liquid Assets and Real Investment: Coefficient plots for IG firms.
Notes: Each point is an estimate of β_{it} from the regression $Y_{it} = \sum_{j=-4}^8 \beta_{it,j} Iss_{it+j} + \alpha_i + \alpha_{it} + \epsilon_{it}$ with 95% confidence intervals. "Cash/2019 assets" is cash and short term investments, normalized by the firm's 2019 year end total assets. "Cash/prior year assets" is normalized by the firm's total assets from one year prior. "PPE/2019 assets" is normalized by the firm's 2019 year end total assets. "PPE/prior year assets" is total property plant and equipment, normalized by the firm's total assets from one year prior. Observations are firm-quarters up to five quarters prior to a bond issuance and eight quarters following a bond issuance, including all issuers rated BBB- or above. "Is" denotes the quarter ending immediately after issuance. We include firm and industry-year fixed effects. Standard errors are clustered by 2-digit industry level. All ratios are winsorized at the 1% level in the entire sample. "Normal" times includes bonds issued between 2010-2019. "Post-Fed" times includes bonds issued March 23-June 30, 2020.

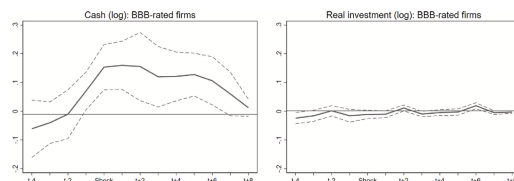


Fig. 2. Second stage estimates: BBB-rated firms.
Notes: Each point is an estimate of β_{it} from the regression Eq. (3), with 95% confidence intervals, where $AbnormalIssued_{it}$ is recovered from the first stage Eq. (2). Cash is cash and short term investments, and Real investment is total plant, property and equipment. Both left-hand-side variables are logged. Includes issuers of credit rating BBB- to BBB+. Observations are firm-quarters up to five quarters before and eight quarters following the Federal Reserve's announcement to intervene in corporate bond markets in March 2020, which is denoted by "Shock". We include firm fixed effects and cluster standard errors by 2-digit industry. Solid horizontal lines are drawn at the value for the coefficient at 2019Q4 (t=-2).

5.2 Figure 2: IV second-stage estimates using Fed index weights

Read.

- Among BBB-rated firms, those with larger policy-induced issuance show significantly larger increases in cash.
- Their investment response is flat: there is no comparable rise in real investment relative to less exposed firms.
- The paper reports similar results for A-rated firms in the appendix.

Economic message. Once the paper moves from raw issuance to instrumented issuance, the same core pattern remains. The intervention increased liquidity accumulation, not average capital spending.

5.3 Figure 3: high-frequency spread exposure and diff-in-diff

Read.

- Firms in the top quartile of policy-driven spread compression accumulate more cash after the announcements.
- With and without quarter fixed effects, the investment response stays weak.
- The appendix reports that treated high-yield firms also do not meaningfully increase investment.

Economic message. The second causal design independently supports the same conclusion: announcement-driven easing in bond financing relaxed balance sheets much more than it stimulated real expenditure.

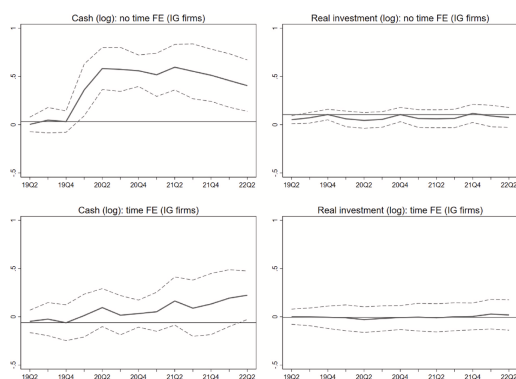


Fig. 3. Policy-driven credit spread change treatment effect and Diff-in-Diff: IG firms.
Notes: Each point is an estimate of β_{it} from the regression $\ln(Y)_{it} = \sum_{q=1}^{2020Q1} (\beta_q \times 1(q = m) \times 1(HighExposure)_{it}) + \alpha_i + (\alpha_q)_{it}$, with 95% confidence intervals, where $1(HighExposure)_{it}$ is a dummy variable that equals 1 if firm i has a predicted policy-driven credit spread reduction recovered from column (3) of Table 1A8 (using only the interaction terms that is in the top 25% of IG firms). Cash is cash and short term investments, and Real investment is total plant, property and equipment. Both left-hand-side variables are logged. Observations are firm-quarters for IG firms from Q1 2019–Q2 2022. The upper panel includes firm fixed effects, the lower panel includes firm and quarter fixed effects, and we cluster standard errors by 2-digit industry. Solid horizontal lines are drawn at the value for the coefficient at 2019Q4.

Table 1
Debt composition: Aggregate flows over 2020Q1.

	HY Billions of USD	IG, BBB Billions of USD	IG, A or above Billions of USD
Bond issuance	111.1	258.4	231.7
Credit line	50.1	73.5	4.04
All bank debt	56.4	102.8	20.0
Undrawn credit EOY 2019	121.2	350.4	152.3

Notes: This table classifies aggregate debt flows based on FISD bond issuance data (Row 1) as well as changes in outstanding debt for other credit instruments during 2020Q1 based on Capital IQ Capital Structure Summary table (Rows 2 and 3). Undrawn credit EOY 2019 is the outstanding available Undrawn Revolving Credit at the end of 2019. Issuers include all U.S. firms that issued a bond March 23–June 30 2020 that we could merge with Capital IQ information.

5.4 Table 1: aggregate debt composition in 2020Q1

Read.

- High-yield issuers raised about **\$111.1B** in bonds while having about **\$121.2B** of undrawn revolving credit at end-2019.
- BBB issuers raised about **\$258.4B** in bonds and had about **\$350.4B** in undrawn revolving credit.
- A-or-above issuers raised about **\$231.7B** in bonds while having about **\$152.3B** in undrawn revolving credit.

Economic message. The firms that rushed into bond markets were not generally firms with no other financing access. Large bond issuers entered the crisis with major committed liquidity from banks.

5.5 Table 2: who actually drew on bank credit lines

Read.

- Among post-March 23 high-yield issuers, **21%** maxed out credit lines, but **32%** did not draw at all; **26%** received no net bank funding.
- Among BBB issuers, only **9%** maxed out, while **48%** did not draw at all and **39%** received no net bank funding.
- Among A-or-above issuers, only **6%** maxed out and **69%** did not draw at all; **66%** received no net bank funding.

Economic message. Even in the middle of the crisis, many issuers left bank liquidity untouched. That makes it harder to view post-announcement bond issuance as pure relief for firms at a binding liquidity corner.

Table 2
Bank borrowing in 2020Q1 for bond issuers.

	HY Share	IG, BBB Share	IG, A or above Share
Maxed out CL (>90%)	0.21	0.09	0.06
High utilization of CL (50%–90%)	0.16	0.09	0.09
Drew some CL (1%–50%)	0.31	0.33	0.26
Did not draw CL (0%)	0.32	0.48	0.69
No net bank funds	0.26	0.39	0.66
Av. drawdown rate	0.41	0.22	0.09

Notes: This table classifies bond issuers based on changes in outstanding debt for different credit instruments during 2020Q1, based on the Capital IQ Capital Structure Summary tables. Row 1 includes issuers that maxed out their credit lines, i.e. the increase in Revolving Credit is at least 90% of Undrawn Revolving Credit at the end of 2019. Row 2 includes issuers that drew some of their credit lines, i.e. the increase in Revolving Credit as a ratio of Undrawn Revolving Credit at the end of 2019 is between 0% and 90%. Row 3 includes issuers that did not draw, i.e. the increase in Revolving Credit is 0 or less. Row 4 includes issuers with no net bank funding, defined as the sum of Revolving Credit, Term Loans and Federal Home Loan Bank borrowings. Row 5 reports the average increase in the drawdown rate, defined as the ratio of Revolving Credit to the Undrawn Revolving Credit at the end of 2019. Includes U.S. firms that issued a bond March 23–June 30, 2020 that we could merge with Capital IQ.

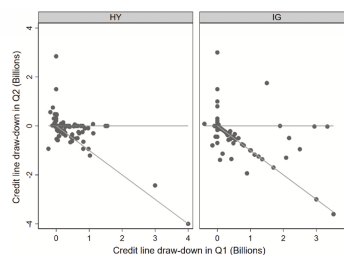


Fig. 4. Loan-bond substitution: Credit line draw-downs in 2020Q2 vs. 2020Q1.
Notes: This figure plots credit line repayment in 2020Q2 against 2020Q1 credit line draw-downs, based on Capital IQ Capital Structure Summary tables, separately for high-yield and investment grade issuers. For ease of interpretation, the figure also displays the negative 45 degree line (exact repayment in Q2) and horizontal line (no change in credit line in Q2). Issuers include all U.S. firms that issued a bond March 23–June 30 2020 that we could merge with Capital IQ information. For clarity, the plots exclude large outliers Volkswagen, Ford, and GM.

5.6 Figure 4: repaying bank loans after bond issuance

Read.

- Many firms that drew in 2020Q1 lie on or near the negative 45-degree line in 2020Q2, meaning they repaid most or all of those drawdowns after issuing bonds.
- Among high-yield issuers that borrowed from banks in Q1, about **74%** repaid some amount and about **42%** fully repaid by end-June.
- Among BBB issuers that drew in Q1, about **91%** repaid some amount after issuance.
- The paper estimates at least **\$125B** of repayment from bond issuers to banks between April and September 2020.

Economic message. A large share of post-announcement bond financing went to *loan-bond substitution*. The intervention did not just increase total liquidity; it also shifted who was providing that liquidity.

5.7 Figure 5: equity repurchases

Read.

- In normal times, bond issuance is often followed by continued equity repurchases.
- In 2020, the probability of repurchase drops by roughly **20 percentage points** after issuance.

- But repurchases do not disappear: about **47%** of issuers still repurchased shares between March and June 2020.

Economic message. There was some precautionary cash preservation, but many issuers still paid out shareholders. That is hard to square with a view that the marginal dollar of liquidity was extremely valuable for all targeted firms.

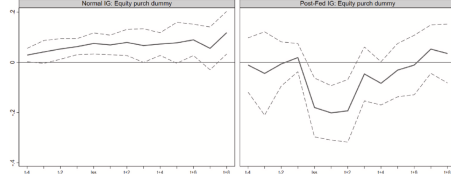


Fig. 5. Equity repurchases: Coefficient plots.
Notes: Each point is an estimate of β_4 from the regression $Y_{it} = \sum_{j=-5}^5 \beta_j \text{Time}_{t-j} + \alpha_i + \alpha_{it} + \epsilon_{it}$ with 95% confidence intervals. “Equity purchase dummy” is an indicator for positive purchases of common or preferred shares in that quarter. Includes investment grade firms (rated BBB- and above). Observations are firm-quarters up to five quarters prior to a bond issuance and eight quarters following a bond issuance. “t=0” denotes the quarter ending immediately after issuance. We include firm and industry-year fixed effects. Standard errors are clustered by 2-digit industry level. All ratios are winsorized at the 1% level. “Normal” times includes bonds issued between 2010-2019, “Post-Fed” times includes bonds issued between March 23-June 30, 2020.

Table 3

Decomposition of \$1 bond issue ($\beta_{t+2} - \beta_{t-2}$)

	Normal: All	Post-Fed: All	Post-Fed: IG	Post-Fed: HY
Rollover bonds	0.48	0.30	0.20	0.43
Pay down bank debt	0.03	0.15	0.09	0.23
Real investment	0.08	-0.00	0.04	-0.06
Payout to equity	0.01	-0.06	-0.09	-0.02
Cash	0.06	0.45	0.61	0.25

Notes: Reports point estimates of $\beta_{t+2} - \beta_{t-2}$ from the regression equation (5). First column includes estimates on the full subset of bonds issued between 2010-2019, second column includes bonds issued between March 23-June 30, 2020, third and fourth columns report estimates on subsamples of bonds issued between March 23-June 30, 2020 by investment grade and high yield issuers, respectively. Total debt is total long term debt plus debt in current liabilities. Senior bonds and notes are from Capital IQ, investment grade firms are rated BBB- and above, and high yield firms are rated below BBB-. Observations are firm-quarters up to five quarters prior to a bond issuance and eight quarters following a bond issuance. “t+2” denotes the quarter end that falls at least one full quarter after bond issuance, and “t-2” denotes quarter end that falls at least one full quarter preceding issuance. We include firm and industry-year fixed effects. Standard errors are clustered by 2-digit industry level. All ratios are winsorized at the 1% level in the entire sample.

5.8 Table 3: decomposition of each dollar of bond issuance

Read.

- In normal times, each dollar of bond issuance is associated with about **48 cents** of bond rollover, **8 cents** of real investment, and only **6 cents** of extra cash.
- Post-Fed, each dollar is associated with about **45 cents** of cash accumulation, **15 cents** of bank-debt repayment, **30 cents** of bond rollover, and approximately **zero** real investment.
- Within post-Fed issuers, investment-grade firms put more into cash (about **61 cents**) while high-yield firms put more into bond rollover (about **43 cents**) and bank-debt repayment (about **23 cents**).

Economic message. This is the paper’s cleanest quantitative punchline. The 2020 issuance boom primarily financed liquid-asset accumulation and liability management, not capital expenditure.

6 Short summary

- The Fed’s 2020 bond-market intervention clearly reopened corporate bond markets.
- Issuers responded by raising enormous amounts of debt, but they used much of it to hold cash, pay down bank loans, and roll over existing bonds.
- Real investment remained weak in both descriptive and causal designs.
- Many issuers already had unused committed liquidity from banks, so the marginal value of extra bond liquidity was often limited.
- Firms that did draw bank lines early in the crisis often used bond proceeds to repay those same banks within a quarter.

- Equity payouts fell somewhat, but they remained widespread enough to show that many issuers were not at a severe internal-funds margin.
- The paper’s broader lesson is that unconventional monetary policy can stabilize funding markets without generating a strong average investment response when targeted firms are already relatively unconstrained.

7 Conclusion

7.1 Core conclusion

- The 2020 corporate bond intervention stimulated issuance very strongly.
- But the firm-level use of funds was dominated by liquidity accumulation and capital-structure adjustment rather than real investment.
- The average real multiplier from bond-market support appears limited in this setting.

7.2 Economic intuition

- Firms value long-term market financing when it becomes temporarily abundant.
- That value is especially strong when firms want to build buffers against future uncertainty or extend debt maturity.
- However, if targeted firms already have access to credit lines and are not the most financially constrained firms in the economy, the extra liquidity need not translate into immediate capital spending.

7.3 Takeaway for macro and policy design

- Models of monetary transmission need explicit roles for cash, long-term debt, and substitution across financing sources.
- Evaluating bond-purchase programs requires looking beyond yields and issuance into firms’ balance sheets, debt structure, and payout choices.
- The paper therefore shifts the question from “Did the Fed reopen the market?” to “Which firms gained liquidity, how much did they value it, and what did they do with it?”